

Homework 7 Solutions

7.1

Problem 1

Prove Case 2 of Theorem 13.2 above.

Solution. Suppose that $\deg(x) \geq 3$. Let a, b, c be three distinct vertices adjacent to x . If any two of a, b, c are adjacent, then we have a clique between those two vertices and x . Otherwise, the vertices a, b, c form an independent set. In either case, we have a clique or an independent set of size 3.

Problem 2

Suppose G is a connected graph. Show that between every pair of distinct vertices of G , there is a simple path connecting x to y in G . (*Hint: consider a shortest path between x and y and use proof by contradiction.*)

Solution. Let x, y be two distinct vertices of G . Since G is connected, there is a path between x and y . Let P be a shortest path between x and y . We claim that P is simple. Suppose for the sake of contradiction that P is not simple. Then there is a vertex in P that is repeated, say v . Let P_1 be the subpath of P from x to the first time v appears, and let P_2 be the subpath of P from the last time v appears to y . Then we can create a new path from x to y by concatenating P_1 and P_2 , which is shorter than P . This contradicts the choice of P as a shortest path. Therefore, every pair of distinct vertices in a connected graph has a simple path connecting them.

Problem 3

The *degree sequence* of a graph is a list of the degrees of its vertices in nonincreasing order. Prove that if a simple graph has degree sequence

$$2, 2, 2, 1, 1, 1, 1$$

then it must be disconnected.

Solution. By the Handshaking lemma, any such graph must have five edges. We showed in class any connected graph on n vertices must have at least $n - 1$ edges. Since this graph has 7 vertices, it must have at least 6 edges to be connected. Therefore, any graph with the given degree sequence must be disconnected.

Problem 4

For $n \geq 2$, define the cycle graph $C_n = (V, E)$ to have

$$V = \{0, 1, \dots, n-1\}$$

and

$$E = \{\{x, y\} \subseteq V : x - y \equiv \pm 1 \pmod{n}\}.$$

Prove that for every $n \geq 2$, the graph C_n is connected.

Solution. Let x, y be two distinct vertices of C_n . We can assume without loss of generality that $x < y$. Then the following is a path from x to y :

$$x \rightarrow x+1 \rightarrow x+2 \rightarrow \dots \rightarrow y-1 \rightarrow y.$$

7.2

Problem 1

Prove Theorem 14.1 above: every graph $G = (V, E)$ is a disjoint union of its connected components, i.e., there are subgraphs $G_1, \dots, G_k$ for some k with disjoint sets of vertices V_i such that each G_i is a connected component of G and

$$\bigcup_{i=1}^k V_i = V.$$

(Hint: induction)

Solution. We prove the statement by induction on the number of vertices of G .

If $|V| = 1$, then G has a single vertex, so it has exactly one connected component, namely G itself. Thus G is a disjoint union of its connected components.

Now assume every graph with n vertices is a disjoint union of its connected components, and let $G = (V, E)$ be a graph with $n+1$ vertices. Choose a vertex $v \in V$, and consider the graph $G-v$ obtained by removing v and all edges incident to it. Since $G-v$ has n vertices, the induction hypothesis implies that $G-v$ is a disjoint union of its connected components. Let these components be $G_1, \dots, G_k$, with vertex sets $V_1, \dots, V_k$. Then the sets $V_1, \dots, V_k$ are pairwise disjoint and

$$V_1 \cup \dots \cup V_k = V \setminus \{v\}.$$

Now add the vertex v back into the graph. If v has no neighbors in $G-v$, then v is an isolated vertex, so $\{v\}$ is a connected component of G . The other connected components remain $G_1, \dots, G_k$.

If v is adjacent to vertices in one or more of the components $G_1, \dots, G_k$, then all components containing a neighbor of v become connected to one another through v , and so together with v they form a single connected component of G . Any component G_i containing no neighbor of v remains a connected component of G .

Therefore the connected components of G still have pairwise disjoint vertex sets, and their union is all of V . Hence G is a disjoint union of its connected components.

By induction, every graph is a disjoint union of its connected components.

Problem 2

The complement of a simple graph $G = (V, E)$ is the graph

$$\overline{G} = (V, \{\{u, v\} : \{u, v\} \text{ is not an element of } E\}),$$

i.e., the graph whose edges are the non-edges of G . Show that for every graph G , if G is not connected then its complement $\overline{G}$ must be connected. (*Hint: consider the connected components of G*)

Solution. Suppose G is not connected. Let $x, y \in V$ be distinct. If x and y are in the same connected component U of G then for any $z \notin U$ then we have $x \rightarrow z \rightarrow y$ is a path in $\overline{G}$. Otherwise if x and y are in different connected components, then $x \rightarrow y$ is a path in $\overline{G}$. Hence $\overline{G}$ is connected.

Problem 3

Let $n \geq 3$ and let C_n denote the cycle graph on n vertices. Prove that C_n is 2-colorable iff n is even.

Solution. Recall that C_n is the graph with vertex set $\{0, 1, \dots, n-1\}$ and edge set $\{\{x, y\} : x - y \equiv \pm 1 \pmod{n}\}$. If n is even, then color the vertices in $\{0, 2, 4, \dots, n-2\}$ with one color and the vertices in $\{1, 3, 5, \dots, n-1\}$ with another color. This is a proper coloring of C_n with two colors. Clearly this is the only way of coloring the graph, as if we color vertex 0 with a color, then we must color vertex 1 with the other color, and so on. Hence if n is odd, clearly this coloring will fail as $n-1$ and 0 will be adjacent and have the same color. Therefore, C_n is 2-colorable if and only if n is even.

Problem 4

Let $n \geq 1$ and define the hypercube graph to be $H_n = (V, E)$ with

$$V = \{(x_1, \dots, x_n) : x_i \in \{0, 1\}\},$$

i.e., the set of bit strings of length n , and

$$E = \{xy : x \text{ and } y \text{ differ in a single bit, i.e., } \exists i \text{ such that } x_i \neq y_i\}.$$

Show that H_n is connected. Show that H_n is 2-colorable.

Solution. Suppose that $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are two distinct vertices of H_n . We can create a path from x to y by changing the bits of x one at a time from left to right until we reach y . For example:

$$x = (0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1) = y.$$

Hence H_n is connected. To see that H_n is 2-colorable, define two sets of vertices:

$$A = \left\{ x \in H_n : \sum_{i=1}^n x_i \text{ is even} \right\}, \quad B = \left\{ x \in H_n : \sum_{i=1}^n x_i \text{ is odd} \right\}$$

Color the vertices in A with one color and the vertices in B with another color. This is a proper coloring of H_n with two colors, as any two adjacent vertices differ in exactly one bit, and changing one bit changes the sum $\sum x_i$ by one. Hence all edges must go from A to B . Hence H_n is 2-colorable.